

Inter-VLAN Routing Basics Theory: Methods of inter-VLAN routing: Router-on-a-stick, Layer 3 switches. Practical: Configure router-on-a-stick for inter-VLAN communication. Task: Troubleshoot inter-VLAN routing issues in a simulated scenario.

Q-1 In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.

[A]

- To create three VLANs representing **Zomato, Flipkart, and Spotify users**, configure **Router-on-a-Stick** for inter-VLAN communication, assign IP addresses and gateways, troubleshoot a failed inter-VLAN connection, and compare Router-on-a-Stick with Layer 3 switching.

○ **Create Three VLANs and Assign PCs**

VLAN ID	VLAN Name	Represents
10	Foodies	Zomato users
20	Shoppers	Flipkart users
30	Streamers	Spotify users

○ **Configure VLANs on the Switch**

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Foodies
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Shoppers
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 30
Switch(config-vlan)# name Streamers Switch(config-vlan)#
exit
```

Q-2 Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

[A]

- Router-on-a-Stick uses **one physical router interface with multiple subinterfaces**, with each subinterface representing a VLAN.

- **Configure Router R1**

```
Router> enable
Router# configure terminal
```

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

- **VLAN 10 — Foodies**

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0 Router(config-
subif)# exit
```

- **VLAN 20 — Shoppers**

```
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0 Router(config-
subif)# exit
```

- **VLAN 30 — Streamers**

```
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
```

- **Configure the Switch Port Connected to Router as a Trunk**

```
Switch# configure terminal
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit

Switch(config)# end
Switch# write memory
```

Q-3 Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

[A]

➤ **PC1 → Desktop → IP Configuration**

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

➤ **PC2 — Shoppers / Flipkart IP Address: 192.168.20.10**

Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

➤ **PC3 — Streamers / Spotify IP Address: 192.168.30.10**

Subnet Mask: 255.255.255.0
Default Gateway: 192.168.30.1

Device	VLAN	IP Address	Default Gateway
PC1 – Foodies	10	192.168.10.10	192.168.10.1
PC2 – Shoppers	20	192.168.20.10	192.168.20.1
PC3 – Streamers	30	192.168.30.10	192.168.30.1

➤ **Test Inter-VLAN Communication**

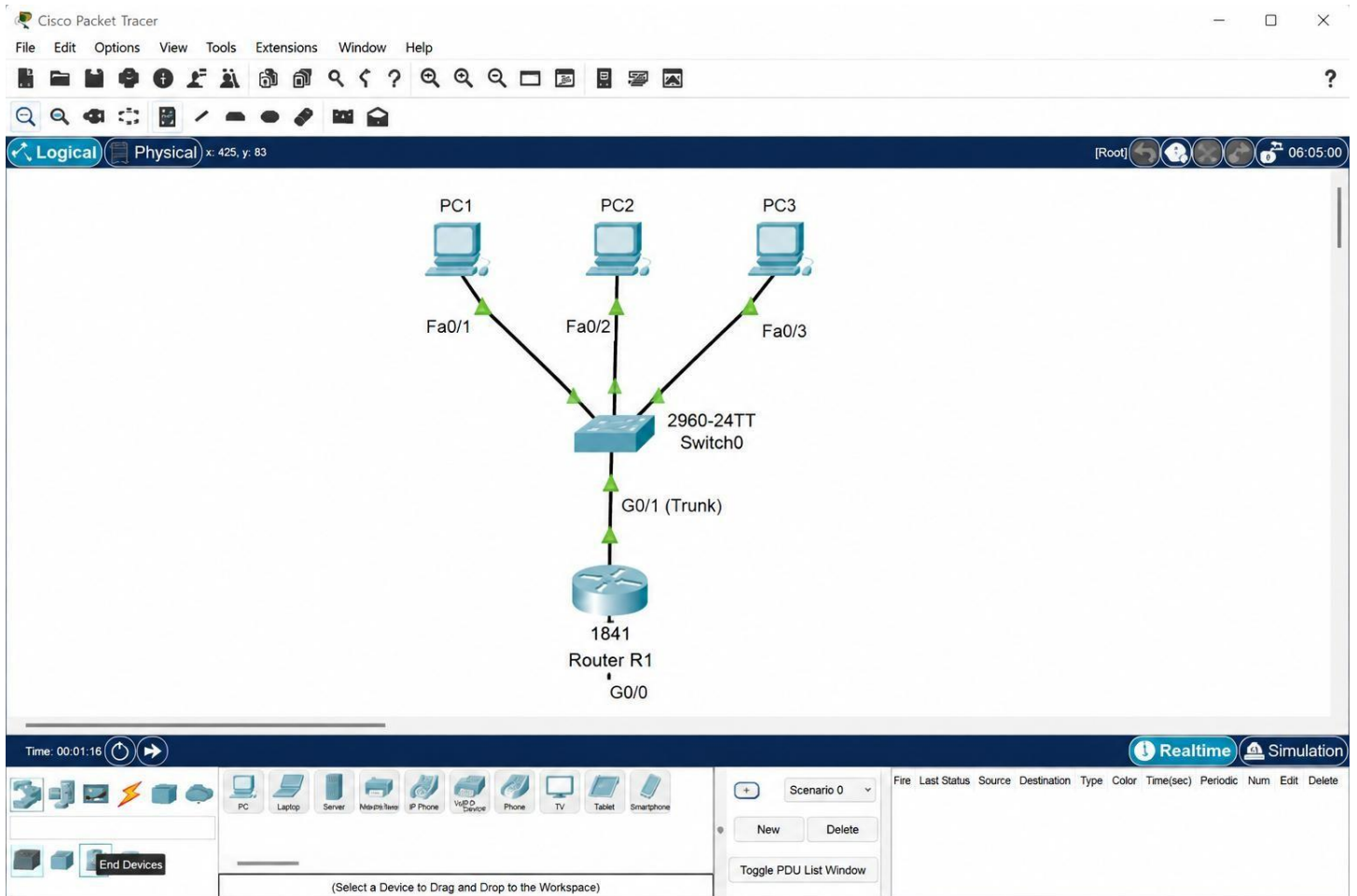
- From **PC1**, open Command Prompt:
ping 192.168.20.10
- Expected:
Reply from 192.168.20.10
- Test PC1 → PC3:
ping 192.168.30.10
- Test PC2 → PC3:
ping 192.168.30.10

Q-4 Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity. Hint: Check VLAN assignments, router sub-interface configurations, and trunk port settings.

[A]

➤ STEPS :

- 1) Step 1: Check VLAN Assignment
- 2) Step 2: Check Router Subinterface
- 3) Step 3: Check the Trunk
- 4) Step 4: Check PC3 Default Gateway
- 5) Step 5: Test the Router Interfaces



Q-5 Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

[A]

Feature	Router-on-a-Stick	Layer 3 Switch
Advantage 1	Less expensive when a router is already available	Faster inter-VLAN routing because routing happens in the switch
Advantage 2	Simple for small networks and labs	Better scalability for large networks with many VLANs

Disadvantage 1	All VLAN traffic uses one physical router link, which can become a bottleneck	Layer 3 switches are generally more expensive
Disadvantage 2	Router or trunk-link failure can affect communication between all VLANs	Configuration and Layer 3 features can be more complex for beginners

➤ Simple Comparison

○ Router-on-a-Stick:

```

VLAN 10 └─┘
VLAN 20 └─┬── Trunk ─ Router
VLAN 30 └─┘
  
```

○ Layer 3 Switch:

```

VLAN 10 └─┘
VLAN 20 └─┬── Layer 3 Switch
VLAN 30 └─┘
  
```